



Reversible quantum circuits for approximate computing with image processing application

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Abstract

This work presents four new architectures of approximate full adders based on reversible quantum circuits, called reversible-logic gate-based approximate full adders (RAFAs). The objective is to develop efficient reversible approximate adders by introducing controlled inaccuracies in the Sum and/or Carry outputs of a 1-bit full adder to achieve resource optimization. This research is motivated by the increasing need for energy-efficient and quantum-compatible arithmetic units, especially for error-tolerant applications such as image processing. In the context of quantum computing, where information loss leads to energy dissipation, reversible logic circuits are essential. However, existing reversible approximate full adders often suffer from high quantum cost (QC), large gate count (GC), and increased resource overhead due to ancillary inputs (AIs) and garbage outputs (GOs). To address these limitations, the proposed RAFAs are evaluated against existing counterparts using key metrics such as GC, QC, AIs, and GOs. Notably, RAFA3 achieves a 50% reduction in QC and a 66.67% reduction in GC compared to the best existing designs. The practical effectiveness of the proposed adders is further validated through image processing (image addition) tasks using image quality metrics, including peak signal-to-noise ratio (PSNR) and structural similarity index metric (SSIM). RAFA4 demonstrates significant improvements, achieving a 7.65% increase in PSNR and a 9.09% improvement in SSIM over the best existing design. The reversible circuits are verified using the Cadence Xcelium tool, and the image processing part is examined using the MATLAB tool.

Keywords Reversible logic · Adders · Approximate computing · Image processing · Quantum computing · Approximate full adders

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1 Introduction

The switching speed and integration density of devices are critical aspects in the evolution of modern computing systems [1]. While Moore's Law has spurred ongoing advancements in silicon-based CMOS technology, this development is reaching physical limits, making circuits smaller than 10 nm more difficult and expensive to build [2, 3]. Moore's Law's survival is increasingly threatened by reliability and fabrication issues. As a result, researchers are looking into alternatives such as More-than-Moore and advanced computational approaches, including quantum and reversible computing for nano-processors [3, 4]. In another aspect, higher integration increases circuit complexity and power demand, raising thermal and reliability concerns. Classical irreversible circuits waste energy and lose information during logic operations—gates like AND, OR, and NOT discard input bits, causing information injury. According to Landauer's principle, this loss leads to energy dissipation as

heat is worsened by dense integration [5, 6]. Bennett showed that reversible logic can reduce such power dissipation [7].

All these reasons play an imperative role in the popularity of reversible computing and quantum computing. Reversible computing involves processes in which no information is lost, allowing the computation to be reversed, resulting in less energy dissipation than irreversible computing [8]. Quantum computing requires reversible operations because quantum gates are unitary and may be reversed [9]. This reversibility is required to maintain quantum coherence and perform complex computations effectively. Both reversible computing and quantum computing seek to minimize energy loss while increasing computational performance by retaining information throughout operations [8, 9]. Thus, reversible computing systems are a viable alternative for minimizing energy dissipation, making them suitable for low-power devices, nano-processors, and quantum computing systems [10].

Reversible computing underpins the theoretical framework of quantum computing. Since quantum operations are inherently reversible, advances in reversible logic design directly support the development of scalable, efficient quantum computers.

These are essential for the development of high-performance, energy-efficient computing in the future. Among the fundamental building blocks of computational systems, the adder plays a crucial role in arithmetic operations.

This work focuses on designing and optimizing reversible adders, which are essential for constructing efficient quantum circuits and enabling practical image processing applications. It is worth noting that reversible adders can be broadly classified into exact and approximate. While exact reversible adders have been more extensively studied, approximate reversible adders—which offer trade-offs between accuracy and resource efficiency—are equally important but have seldom been explored in the literature. This gap highlights a pressing need for further research, which serves as the main motivation for our work.

This work presents new approximate full adders utilizing reversible logic, referred to as reversible logic approximate full adders (RAFAs), planned explicitly for image processing applications. The suggested RAFAs are evaluated in comparison to state-of-the-art architectures considering essential design criteria, such as gate count (GC), auxiliary inputs (AIs), quantum cost (QC), and garbage outputs (GOs). This work presents the application in image processing, i.e. an image addition executed in MATLAB to evaluate its effectiveness in image processing.

The primary goal of this study is to develop efficient hardware circuits for error-tolerant applications by combining the strengths of reversible logic and approximate computing. Reversible logic ensures low-energy operation,

while approximate computing enables hardware-efficient circuit design.

The main contributions are:

1. Design approximate adders with fewer logic gates and fewer constant inputs.
2. Examine the practical application of the suggested adders using image processing.
3. A rigorous parameter analysis, including cost, is done to propose better designs.

The structure of this paper is as follows: After the introduction in Sect. 1, Sect. 2 provides an overview 2. Foundations of Reversible logic. Section 3 details the proposed reversible approximate full adders (RAFAs) logic. Section 4 discusses the application of RAFAs in image processing. The comparison with existing works is included in Sect. 5. The final section is Sect. 6, which presents the conclusions drawn from this work.

2 Foundations of reversible logic

This section discusses the fundamentals of reversible quantum circuits with exact and approximate reversible adders, which are necessary to understand the upcoming sections.

2.1 Reversible logic and quantum computing

Reversible logic operations offer a highly promising approach in computing, as they preserve all input information without discarding any data. For a system to be considered reversible, it must allow backward computation—that is, its outputs must uniquely determine the original inputs. Such systems are often described as lossless circuits because they eliminate both information loss and energy dissipation. These characteristics make reversible circuits particularly attractive for applications requiring ultra-low power consumption and minimal heat generation, such as communication systems, low-power VLSI design, nano computing, and nanotechnology.

A reversible circuit is composed of fundamental reversible gates. Reversible logic is essential in quantum computing, where information is represented by qubits. Qubits, unlike standard bits, can be in either state 0, 1, or any quantum superposition of these states. Several reversible gates are central to quantum computing, including the NOT gate (X gate) [11], Controlled-NOT (CNOT or Feynman gate) [12], Controlled-V and Controlled-V⁺ gates [13], Peres gate [14, 15], Fredkin gate [14, 15], Toffoli gate (CCNOT) [14, 15], Double Feynman Gate (DFG) [14, 15], and a special gate, known as Thapliyal–Srinivas gate (TS3G) [16]. These

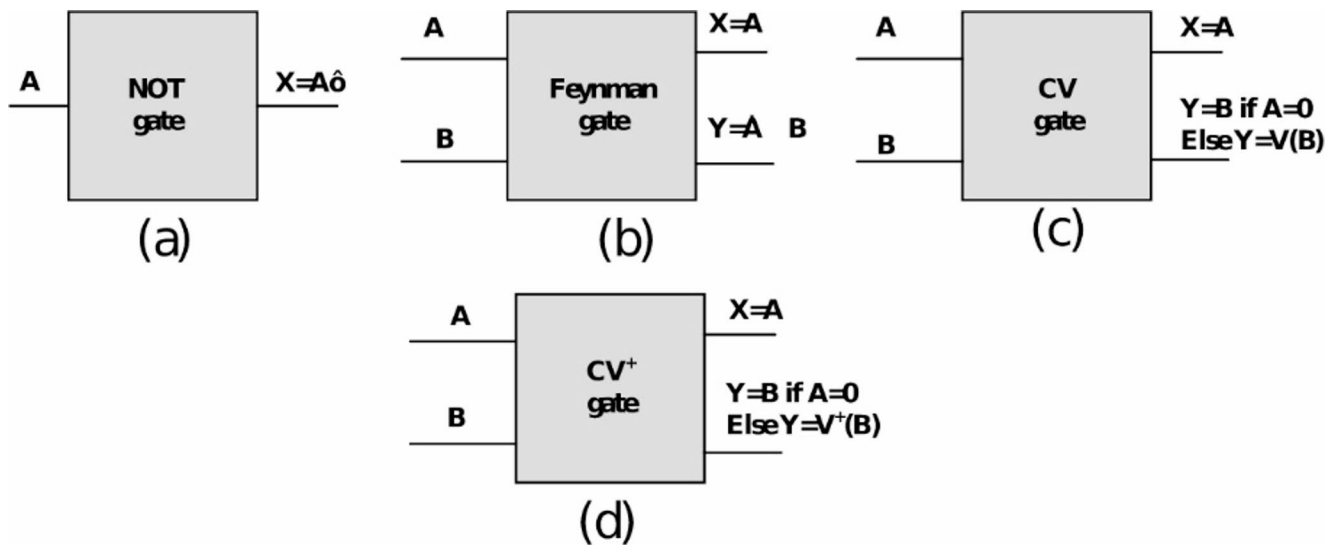


Fig. 1 Standard 1×1 and 2×2 reversible logic gates

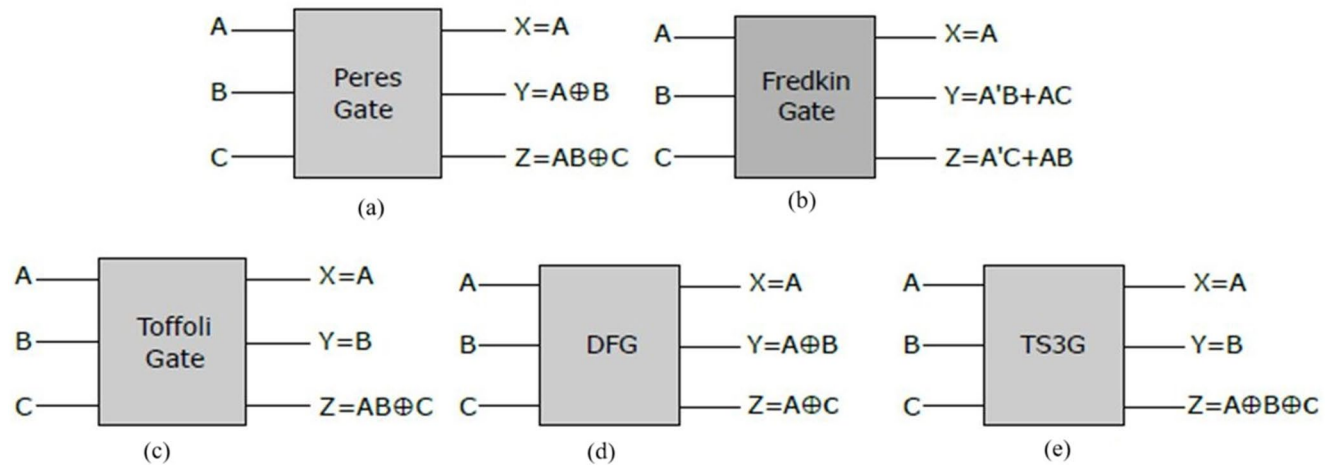


Fig. 2 Standard 3×3 reversible logic gates

Table 1 Size and QC of various rlg

RLG	Size	QC	Work
NOT gate	1×1	1	[11]
CNOT gate	2×2	1	[12]
CV gate	2×2	1	[13]
CV ⁺ gate	2×2	1	[13]
CCNOT gate	3×3	5	[14]
Peres Gate	3×3	4	[14]
Fredkin Gate	3×3	5	[15]
DFG	3×3	2	[15]
TS3G	3×3	3	[16]

gates, illustrated in Figs. 1 and 2, serve as fundamental building blocks for constructing reversible circuits, which are becoming increasingly important in the design of low-power computing devices, quantum computing systems, and other advanced technologies, as discussed in the previous

section. The size and quantum cost (QC) of these gates are shown in Table 1 below.

2.1.1 1×1 reversible block

NOT gate (X gate): The NOT gate, illustrated in Fig. 1(a) and commonly referred to as the Pauli-X gate, inverts the state of a qubit. Given an input A , it produces an output of

$Y = \bar{A}$. This gate is essential in quantum computing, as it carries out a fundamental bit-flip operation. Its quantum cost is 1, and it is a 1×1 logic gate [11].

2.1.2 2×2 reversible blocks

Controlled-NOT (CNOT) gate or Feynman gate: The CNOT gate functions with two qubits: one acting as the control

and the other as the target. As depicted in Fig. 1(b), it takes inputs A and B and produces outputs X and Y accordingly. It is a 2×2 gate, where $X=A$ and $Y=A \oplus B$. It is also called the quantum XOR gate. The quantum cost of the gate is 1 [12].

Controlled- V gate (CV gate): A control qubit in A and a target qubit in B of Controlled- V Gate (CV gate) producing outputs X and Y . It is a 2×2 block, as shown in Fig. 1(c). The control qubit remains unchanged, so $X=A$. The target output Y depends on the state of the control qubit, like if $A=0$, then $Y=B$; $A=1$, then $Y=V(B)$, where V is the square root of the NOT gate. The V gate is reversible, and the controlled version allows for partial state changes conditional on the control qubit's state. Its quantum cost is 1, and the Controlled- V gate is a fundamental building block in quantum circuits [13].

Controlled V^+ gate (CV^+ gate): The Controlled V^+ gate or CV^+ gate functions on two qubits. A control qubit in A and a target qubit in B producing outputs X and Y . It is a 2×2 quantum gate, as shown in Fig. 1(d). The control qubit remains unchanged, so $X=A$. The target output Y depends on the state of the control qubit, like if $A=0$, then $Y=B$; $A=1$, then $Y=V^+(B)$, where V^+ is the inverse of the V gate. Its quantum cost is 1 [13].

2.1.3 3×3 reversible blocks

Toffoli gate (CCNOT): The Controlled-Controlled-NOT gate (CCNOT) gate, also known as the Toffoli gate (shown

in Fig. 3(c)), is a 3-input, 3-output reversible logic gate that performs a NOT operation on the third input (target qubit) only when the first two inputs (control qubits) are both set to 1. The quantum cost of the CCNOT gate is typically 5, as it requires a series of simpler gates to implement the controlled operation [14, 15].

Peres gate: The Peres gate is a 3×3 reversible logic gate, depicted in Fig. 2(a). It takes three inputs (A, B, C) and produces outputs $P=A$, $Q=A \oplus B$, and $R=AB \oplus C$, effectively combining the functions of a Feynman gate and a Toffoli gate. This makes it particularly useful in arithmetic circuits, such as reversible adders. The Peres gate is associated with a quantum cost of 4 [14, 15].

Fredkin gate: The Fredkin gate performs a controlled switch operation, shown in Fig. 2(b). Its size is 3×3 . It takes three inputs (A, B, C) and produces outputs $P=A$, $Q=A'B+AC$, and $R=A'C+AB$, meaning that the gate passes the control input A unchanged and conditionally swaps the other two inputs (B and C) only if $A=1$. This conservative gate, which preserves the number of logical 1s, is particularly useful in reversible circuit designs where data integrity and energy efficiency are crucial. The quantum cost of the Fredkin gate is typically 5 [14, 15].

Double feynman gate (DFG): The Double Feynman Gate (DFG) is a 3×3 reversible logic gate that performs two simultaneous controlled-NOT (CNOT) operations, making it efficient for reversible circuit design as shown in Fig. 2(d). It accepts three inputs (A, B, C) and produces outputs $P=A$, $Q=A \oplus B$, and $R=A \oplus C$, where the first input A

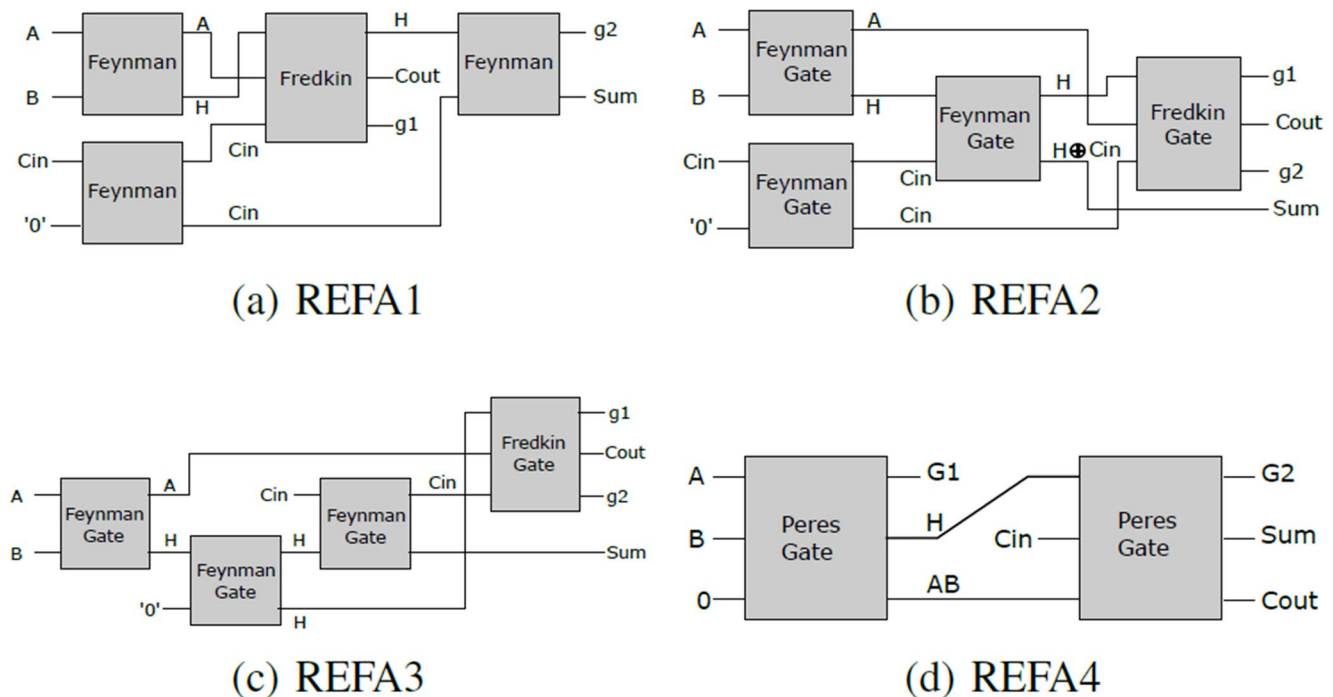


Fig. 3 RLG-based exact full adders (REFAs)

controls both XOR operations. With a quantum cost of 2, the DFG is more cost-effective than implementing two separate Feynman gates [14, 15].

Thapliyal-Srinivas gate (TS3G): It is a specialized gate that plays a significant role in implementing reversible circuits. Thapliyal-Srinivas Gate or TS3G (3×3) is a special implementation of TSG gate, suggested by Thapliyal & Srinivas in [16]. The TS-3 gate can realize any Boolean function, as its third output effectively performs the operation of a three-input XOR gate. Specifically, the outputs are $X=A$, $Y=B$, and $Z=A \oplus B \oplus C$ for inputs A , B , and C . Moreover, the gate helps address the fan-out issue by directly passing two inputs to the outputs [16].

2.2 Reversible exact full adders

The full adder is a fundamental part of arithmetic circuits and significantly influences the performance of the system in which it is used. This is because the full adder is responsible for carrying out basic binary addition, handling both the sum and carry operations. The full adder's efficiency, speed, and power consumption directly influence the system's arithmetic operations in complex digital systems, such as microprocessors, digital signal processors, and arithmetic logic units (ALUs). Thus, optimizing the design and performance of the full adder is essential for improving the entire system's speed, power efficiency, and overall reliability. The performance of a system is greatly influenced by factors such as propagation delay, area efficiency, and power dissipation in the full adder, making it a key component in high-performance, higher-order logic circuits.

The efficacy of reversible logic-based exact full adders (REFAs) is assessed according to certain critical design parameters, each of which significantly impacts the overall efficiency and practicality of the circuit.

These metrics include gate count (GC), quantum cost (QC), constant or ancillary inputs (AIs), and garbage outputs (GOs). Together, these metrics provide a comprehensive view of reversible full adders' efficiency, speed, and

resource consumption, helping designers optimize circuits for various applications such as quantum computing, low-power devices, and energy-efficient computing systems.

In [17], an innovative 1-bit full adder architecture (REFA1) was introduced with a quantum cost of 8, as shown in Fig. 3(a). An adiabatic exact full adder architecture (REFA2) was proposed in [18], offering reduced delay while maintaining the same quantum cost of 8, as shown in Fig. 3(b). A carry-dependent reversible full adder (REFA3) was reported in [19], also with a quantum cost of 8, as shown in Fig. 3(c). In [20], the proposed reversible logic full adder (REFA4) demonstrated low power loss, with a QC of 8, as depicted in Fig. 3(d). All the discussed adders are reversible, and they are exact full adders, as shown in Fig. 3. These adders are designed using the standard RLGs as discussed in the earlier section.

2.3 Reversible approximate full adders

This section introduces the architectures of previously reported approximate full adders based on reversible logic. In recent literature, only two reversible approximate full adders (RAFAs) have been documented, specifically RAFA1 and RAFA2 [21]. Figure 4 illustrates the schematic of these two approximate adders, and their respective truth tables are provided in Table 2.

The output equations for RAFA1 and RAFA2 in the input signals A , B , and C_{in} are given in Eqs. (1) and (2), respectively. Both designs are derived by approximating the carry output (C_{out}) while preserving a conventional full adder's exact sum (F_{Sum}). The schematic of RAFA1 is shown in Fig. 4(a). It is implemented using three Feynman gates with a total quantum cost (QC) of 3. RAFA1 has one constant input and two garbage outputs labelled $g1$ and $g2$. The schematic of RAFA2 is shown in Fig. 4(b). This design uses two Feynman gates, with a QC of 2, and has one garbage output, $g1$.

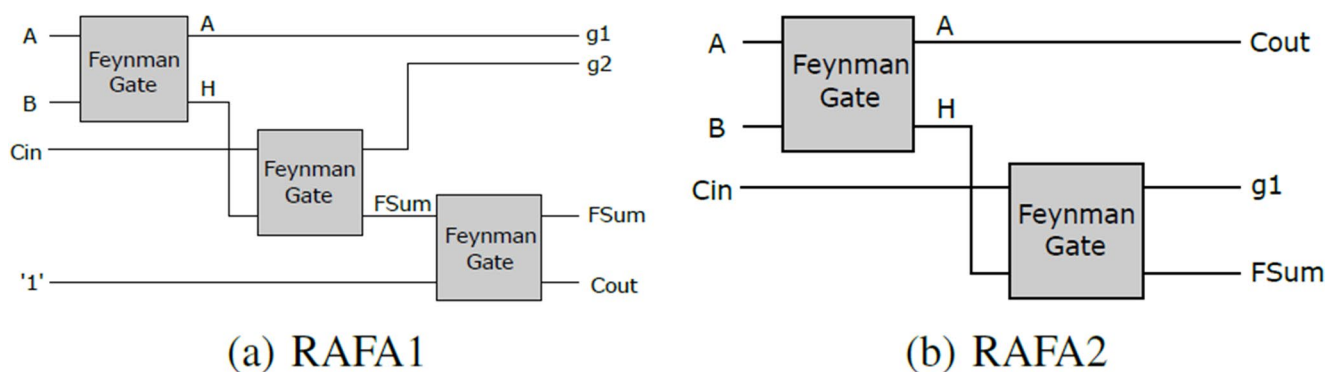


Fig. 4 Existing reversible logic approximate adders (RAFA1 & RAFA2)

Table 2 Truth table of exact and approximate full adders

Inputs			EFA			RAFA1			RAFA2			RAFA3			RAFA4			RAFA5			RAFA6		
A	B	C _{in}	Cout	Sum	ED	Cout	Sum	ED	Cout	Sum	ED	Cout	Sum	ED	Cout	Sum	ED	Cout	Sum	ED	Cout	Sum	ED
0	0	0	0	0	2	0✓	0✓	0	0✓	0✓	0	0✓	1×	1	0✓	0✓	0	0✓	0✓	0	0✓	0✓	0
0	0	1	0	1	0	0✓	1✓	0	0✓	1✓	0	0✓	1✓	0	1×	0×	0	0✓	1✓	0	0✓	1✓	0
0	1	0	0	1	0	0✓	1✓	0	0✓	1✓	0	0✓	1✓	0	0✓	1✓	0	0✓	1✓	0	0✓	0×	-1
0	1	1	1	0	0	0✓	0✓	0	0×	0✓	-2	0×	1×	-1	1✓	1×	1	0×	0✓	-2	1✓	0✓	0
1	0	0	0	1	0	0✓	1✓	0	1×	1✓	2	1×	0×	1	0✓	0×	-1	1×	1✓	2	0✓	1✓	0
1	0	1	1	0	0	0✓	0✓	0	1✓	0✓	0	1✓	0✓	0	1✓	0✓	0	1✓	0✓	0	1✓	1×	1
1	1	0	1	0	0	0✓	0✓	0	1✓	0✓	0	1✓	0✓	0	0×	1×	-1	1✓	0✓	0	1✓	0✓	0
1	1	1	1	1	-2	1✓	1✓	-2	1✓	1✓	0	1✓	0×	-1	1✓	1✓	0	1✓	1✓	0	1✓	1✓	0

$$\left. \begin{aligned} Sum &= A \oplus B \oplus C_{in} \\ C_{out} &= A \oplus B \oplus C_{in} \end{aligned} \right\} \quad (1)$$

$$\left. \begin{aligned} Sum &= A \oplus B \oplus C_{in} \\ C_{out} &= A \end{aligned} \right\} \quad (2)$$

The primary gaps in the existing designs are the increased number of gates and higher constant inputs. The aim of this work is to design a reversible adder using fewer constant inputs and logic gates. This work also illustrates the practical applications of these adders in image processing.

3 Proposed reversible approximate full adders

This section presents four novel reversible logic-based approximate full adders referred to as RAFA3, RAFA4, RAFA5, and RAFA6 and examines their advantages over existing reversible approximate full adders, using reversible logic design metrics for evaluation, as shown in Fig. 5. The reversible circuits are verified using Cadence Xcelium tool (version 23.09). The truth table of these adders and error distance (ED), are already mentioned in Table 2. In this table, ED represents the maximum error deviation from the expected output. Both the proposed designs and the existing architectures exhibit symmetrical deviations in positive and negative directions. The key features of these adders are summarized in Table 3 and outlined as follows:

The schematic of proposed RAFA3 is shown in Fig. 5(a). This design is based on the concept of approximating both the sum and carry-out (C_{out}) of a conventional full adder, with the output equation presented in Eq. (3). RAFA3 is implemented using a single 2×2 Feynman gate (Gate Count, GC=1), resulting in a total quantum cost (QC) of 1 and featuring one constant input. The primary advantage of this adder is its minimal QC and the absence of garbage output (GO).

$$\left. \begin{aligned} Sum &= \bar{A} \\ C_{out} &= A \end{aligned} \right\} \quad (3)$$

Figure 5(b) presents the schematic of the proposed RAFA4, designed using a single 3×3 Double Feynman gate. This design approximates both the sum and the carry-out (C_{out}) of a conventional full adder, with the corresponding equations provided in Eq. (4). The key metrics for this adder include a QC of 2, one garbage output (GO), and one constant input (AI).

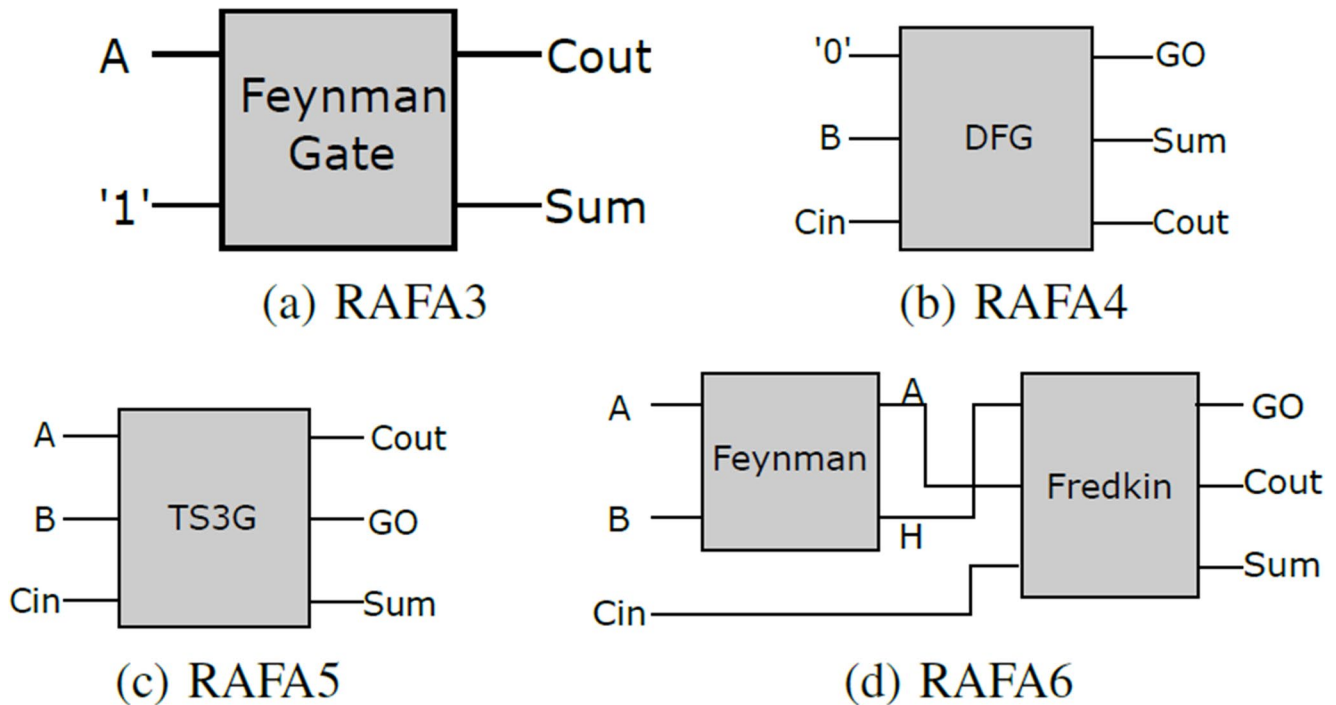


Fig. 5 Proposed reversible logic approximate adders (RAFA3- RAFA6)

Table 3 Proposed and existing rafas in terms of GC, AIS, GOS, AND QC

Adders	Approximation type	GC	AIs	GOs	QC	Works
RAFA1	Exact Sum and Approximate Carry	3	1	2	3	[21]
RAFA2	Exact Sum and Approximate Carry	2	0	1	2	[21]
Proposed RAFA3	Approximate Sum and Approximate Carry	1	1	0	1	This work
Proposed RAFA4	Approximate Sum and Approximate Carry	1	1	1	2	This work
Proposed RAFA5	Exact Sum and Approximate Carry	1	0	1	3	This work
Proposed RAFA6	Approximate Sum and Exact Carry	2	0	1	6	This work

$$\left. \begin{aligned} Sum &= B \\ C_{out} &= C_{in} \end{aligned} \right\} \quad (4)$$

Figure 5(c) illustrates the schematic of the proposed RAFA5, designed using a single 3×3 TS3 gate (TS3G). This design retains the exact sum while approximating the Cout of a conventional full adder, with the output equations presented in Eq. (5). The key features of this adder include QC=3, AI=0, and GO=1.

$$\left. \begin{aligned} Sum &= A \oplus B \oplus C_{in} \\ C_{out} &= A \end{aligned} \right\} \quad (5)$$

Figure 5(d) illustrates the schematic of the proposed RAFA6, which is designed using one Feynman gate and one Fredkin gate. This design approximates the sum while retaining a conventional full adder's exact carry-out (Cout), with the output equations presented in Eq. (6). The key features of this adder include QC=6, AI=0, and GO=1.

$$\left. \begin{aligned} H &= A \oplus B \\ Sum &= HA + \bar{H} C_{in} \\ C_{out} &= \bar{H} C_{in} \end{aligned} \right\} \quad (6)$$

4 Image processing applications

This section evaluates the suitability of the proposed RAFAs for image processing applications through a simple image addition example. The MATLAB tool (version R2023a) examines all the image processing operations. We utilize two grey-scale images in PNG format, each sized 256×256 pixels, totalling 65,536 pixels. The two test images, a circle (Test image1) and a cameraman (Test image2)-exhibit contrasting features and are used for the addition operation. To perform the addition, we design and simulate six different combinations of ripple carry adders (RCAs) using MATLAB. Each RCA consists of four most significant bit (MSB) adders and four least significant bit (LSB) adders,

Fig. 6 Image Processing of reversible logic-based approximate adders

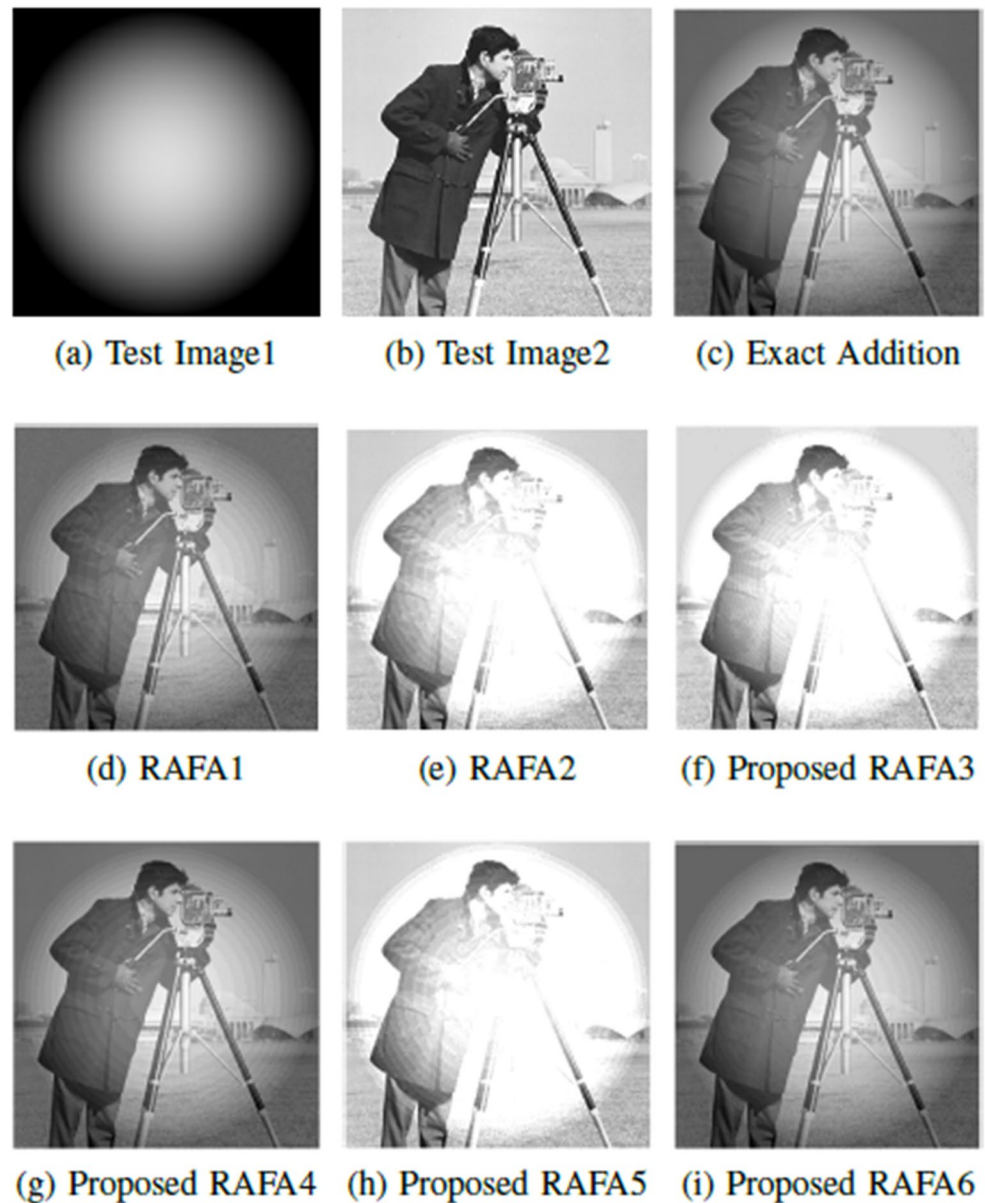


Table 4 Comparisons of proposed and existing rafas in terms of image processing metrics and QC

Adders	PSNR	SSIM	QC
RAFA1 [21]	31.77	0.88	44
RAFA2 [21]	8.10	0.59	40
Proposed RAFA3	8.00	0.58	36
Proposed RAFA4	34.20	0.96	40
Proposed RAFA5	8.10	0.59	44
Proposed RAFA6	34.93	0.94	56

with the MSB and LSB adders implemented using exact and approximate reversible logic-based full adders. The resulting image from the addition is shown in Fig. 6. Additionally, the design metrics of the resulting images, including PSNR, SSIM and QC (for 8-bit RCA), are summarized in Table 4.

To evaluate and compare the performance of the RAFAs, we estimated the Figures of Merit (FoM)=QC ×SSIM for 8-bit ripple carry adders designed using various RAFAs. The results are shown in Fig. 6, from which the following observations can be made:

- RAFA3 and RAFA5: These designs are optimized for QC but with reduced accuracy. The lower QC is attributed to their specific architectural configurations.
- RAFA4: This design achieves the best balance, offering optimal performance in both SSIM and QC compared to the other RAFAs.
- RAFA6: While achieving the highest SSIM among the designs, RAFA6 does so at the expense of a higher QC.

These results highlight the trade-offs in design priorities among the RAFAs, where certain architectures favour QC, others prioritize SSIM, and some aim to optimize both.

5 Comparisons

Table 3 compares proposed and existing reversible approximate full adders (RAFAs) based on their approximation type, gate count (GC), approximate inputs (AIs), garbage outputs (GOs), and quantum cost (QC). RAFA1 and RAFA2, from previous works, employ exact sums with approximate carry outputs, with RAFA2 achieving better efficiency due to fewer GOs and a lower QC—the proposed RAFAs present varied designs with improved performance metrics. RAFA3 and RAFA4 use approximate sums and carries, achieving the lowest GC of 1 and QC values of 1 and 2, respectively. RAFA5 balances an exact sum with an approximate carry while maintaining a low GC and a moderate QC of 3. RAFA6, which combines an approximate sum with an exact carry, has a higher QC of 6. The proposed designs generally offer improved efficiency, particularly RAFA3 and RAFA4. The proposed RAFAs demonstrate notable improvements in gate count (GC) and quantum cost (QC) compared to the existing designs in [21]. RAFA3 achieves the most significant enhancement, reducing both GC and QC by approximately 66.67% compared to RAFA1 and 50% compared to RAFA2. RAFA4 also shows considerable improvement, with a 66.67% reduction in GC and a 33.33% reduction in QC compared to RAFA1 while maintaining the same QC as RAFA2. RAFA5 offers a 66.67% reduction in GC compared to RAFA1 and a 50% reduction in GC compared to RAFA2. The RAFA6 achieves only a modest 33.33% reduction in GC relative to RAFA1, while its QC increases substantially, making it less efficient in terms of quantum cost. Overall, RAFA3 and RAFA4 stand out as the most efficient designs, achieving the highest reduction in both GC and QC.

Table 4 highlights that RAFA4 and RAFA6 achieve the best image quality, with RAFA6 having the highest PSNR of 34.93 and RAFA4 achieving the highest SSIM of 0.96. RAFA1 also shows good image quality but with a higher QC. In contrast, RAFA2, RAFA3, and RAFA5 have lower PSNR and SSIM values, indicating weaker image quality. Among the proposed designs, RAFA3 offers the lowest QC, while RAFA6 balances improved image quality with a higher QC. The proposed RAFAs demonstrate varying degrees of improvement in image processing metrics and quantum cost compared to the existing designs in [21]. RAFA3 shows a 10% reduction in quantum cost compared to RAFA2, though it exhibits slight decreases of approximately 1.23% in PSNR and 1.69% in SSIM. RAFA4 achieves notable improvements, with a 7.65% increase in

PSNR, a 9.09% improvement in SSIM, and a 9.09% reduction in quantum cost compared to RAFA1. RAFA6 delivers the highest PSNR improvement of 9.94% and a 6.82% gain in SSIM; however, this comes at the cost of a 27.27% increase in quantum cost. RAFA5 shows no improvement in quantum cost and demonstrates reduced image quality compared to RAFA1. Overall, RAFA4 emerges as the most balanced design, combining improved image quality with reduced quantum cost. The FoM of different RAFAs are depicted in Fig. 7.

6 Conclusion

The primary goal of this work was to design efficient hardware circuits for error-tolerant applications by leveraging the advantages of reversible logic and approximate computing. Reversible logic facilitates low-energy operation, whereas approximate computing promotes hardware-efficient circuit design. This research introduced four innovative approximate adders utilizing quantum reversible logic gates. The new designs were rigorously assessed against current equivalents utilizing essential quantum design criteria, such as gate count, constant (ancilla) inputs (AI), garbage outputs (GO), and quantum cost (QC). Their practical application was illustrated through the installation of 8-bit ripple carry adders (RCAs), enabling picture addition with both the proposed and current reversible approximate adders. Additionally, the efficacy of each design was measured using the figure of merit (FoM). The examination of FoM reveals that the suggested adders offer a versatile compromise between quality of compression (QC) and the structural similarity index measure (SSIM). Among these reversible approximate full adders (RAFAs), some architectures are optimized for minimal quantum cost (QC), others emphasize structural similarity index measure (SSIM), and certain designs attain a balanced optimization of both metrics. This versatility facilitates customized design decisions according to particular application needs, guaranteeing a flexible methodology for quantum approximation computation. As a note, the reversible circuits are validated in the Cadence Xcelium platform, while the image processing component is analyzed using MATLAB software.

The proposed designs possess specific constraints. The complexity of the hardware escalates with the quantity of adder bits, resulting in a corresponding increase in garbage outputs and constant inputs. Furthermore, these adders are appropriate solely for applications where precise calculation is not essential, such as image and video processing. Notwithstanding these limitations, the suggested RAFAs present a viable avenue for the efficient and adaptable design of quantum arithmetic.

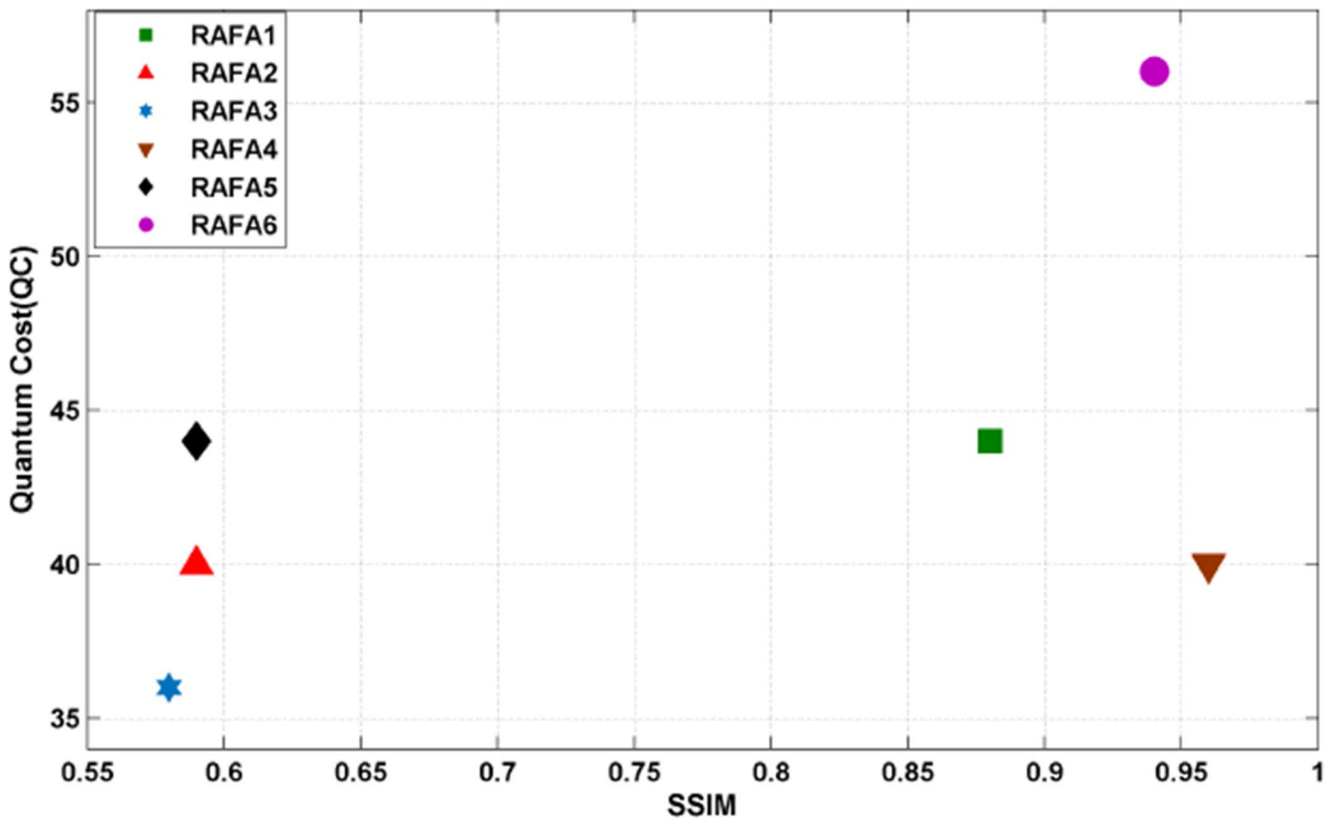


Fig. 7 Figure-of-Merit (FoM) for 8-bit ripple carry adders designed using proposed and existing RAFAs

Future endeavours should optimize designs to save hardware overhead while preserving computational accuracy and enhancing their application. Additional investigation into error-resilient quantum circuits and hybrid classical-quantum approximation computing methods may improve their practical applicability.

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Declarations

Competing interests The authors declare no competing interests.

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